Instead of constructing the triangle, we can also find the value of $\cos\theta$ using $\cos\theta = \pm\sqrt{1-\sin^2\theta}$ formula.

Example 3.24 Prove that $\sin 4A = 4\sin A\cos^3 A - 4\cos A\sin^3 A$

Solution:

$$\begin{aligned} 4\sin A\cos^3 A - 4\cos A\sin^3 A &= 4\sin A\cos A\left(\cos^2 A - \sin^2 A\right) \\ &= 4\sin A\cos A\cos 2A = 2(2\sin A\cos A)\cos 2A \\ &= 2(\sin 2A)\cos 2A = \sin 4A. \end{aligned}$$

Example 3.25 Prove that $\sin x = 2^{10}\sin\left(\frac{x}{2^{10}}\right)\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{2^2}\right)\ldots\cos\left(\frac{x}{2^{10}}\right)$

Solution:

$$\begin{aligned} \sin x &= 2\sin\frac{x}{2}\cos\frac{x}{2} = 2\times 2\times\sin\frac{x}{2^2}\cos\frac{x}{2^2}\cos\frac{x}{2} \\ &= 2^2\sin\frac{x}{2^2}\cos\frac{x}{2}\cos\frac{x}{2^2} \end{aligned}$$

Applying repeatedly the half angle sine formula, we get

$$\sin x = 2^{10}\sin\left(\frac{x}{2^{10}}\right)\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{2^2}\right)\ldots\cos\left(\frac{x}{2^{10}}\right).$$

The above result can be extended to any finite number of times.

Example 3.26 Prove that $\dfrac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta} = \tan\theta$

Solution:

We have $\dfrac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta} = \dfrac{\sin\theta+2\sin\theta\cos\theta}{\cos\theta+(1+\cos 2\theta)} = \dfrac{\sin\theta(1+2\cos\theta)}{\cos\theta(1+2\cos\theta)} = \tan\theta.$

Example 3.27 Prove that $1-\dfrac{1}{2}\sin 2x = \dfrac{\sin^3 x+\cos^3 x}{\sin x+\cos x}$

Solution:

We have $$\begin{aligned} \frac{\sin^3 x+\cos^3 x}{\sin x+\cos x} &= \frac{(\sin x+\cos x)\left(\sin^2 x-\sin x\cos x+\cos^2 x\right)}{\sin x+\cos x} \\ &= 1-\sin x\cos x = 1-\frac{1}{2}\sin 2x. \end{aligned}$$